Zero-Shot vs Few-Shot Prompting for Violence Detection using Large Language Models on Low Resource Language (Bangla).

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ACM Reference Format:
Sourav Saha. 2026. Zero-Shot vs Few-Shot Prompting for Violence Detection using Large Language Models on Low Resource Language (Bangla).. In . https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Abstract

Detecting violence-inciting content on social media is critical for preventing real-world harm, particularly in low-resource languages like Bangla. While prior work has benchmarked traditional NLP methods and fine-tuned BERT models on the Vio-Lens dataset [**?**], the capabilities of large language models (LLMs) in this setting remain unexplored. This project evaluates LLM capability under zero-shot and few-shot prompting conditions on the Vio-Lens benchmark, classifying Bangla social media posts into Direct Violence, Passive Violence, and Non-Violence. Zero-shot GPT-4o achieves a macro F1 of 0.6559, surpassing all n-gram SVM baselines without any training data. Few-shot prompting further improves performance to a macro F1 of 0.7202, exceeding fine-tuned MBERT and approaching XLM-RoBERTa, demonstrating the strong potential of LLMs for low-resource violence detection.

2 Introduction & Problem Statement

Social media platforms have become vehicles for inciting communal violence, particularly in South Asia. In Bangladesh and West Bengal, violence-inciting Bangla content has been directly linked to real-world incidents causing loss of life and property [7]. Automated detection of such content is therefore a critical safety problem.

Despite progress in hate speech detection for high-resource languages [4, 8], Bangla remains severely underrepresented in NLP research. The Vio-Lens dataset [7] addresses this gap with 6,046 annotated social media posts, with BanglaBERT achieving the best macro F1 of 0.7879 on the validation set using full fine-tuning.

The emergence of powerful LLMs raises a compelling question: *can LLMs match fine-tuned models on low-resource violence detection without task-specific training?* This is especially relevant given the high cost of annotating data and fine-tuning models for every new language. This project investigates: (1) the effectiveness of zero-shot LLM prompting for Bangla violence detection; (2) whether few-shot examples significantly improve performance; and (3) how these results compare to established baselines from prior work [7].

Conference'17, July 2017, Washington, DC, USA
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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

3 Related Work

Violence and Hate Speech Detection. Prior work on online hate speech detection has primarily focused on high-resource languages [4, 8]. Bangla-specific efforts include BD-SHS [6] for hate speech and SentNob [3] for sentiment analysis on noisy text. Vio-Lens [7] is the first dataset targeting communal violence incitement in Bangla, with an annotation framework grounded in Galtung's theory of violence [2].

LLMs for Classification. GPT-4 and instruction-tuned LLMs have demonstrated strong zero-shot and few-shot performance across diverse NLP tasks [5]. However, their effectiveness on low-resource languages and sensitive tasks such as violence detection remains largely unexplored. Our work directly addresses this gap by situating LLM performance within an established benchmark.

Prompting Strategies. Zero-shot prompting relies solely on task descriptions, while few-shot prompting provides labeled examples in-context [1]. The latter has been shown to substantially improve performance on classification tasks, motivating our comparison of both approaches on the same test conditions.

4 Methodology

4.1 Dataset

We use the Vio-Lens dataset [7], comprising 6,046 annotated Bangla YouTube comments labeled as **Direct Violence** (explicit calls for killing, attack, or deportation), **Passive Violence** (abuse, insults, or justification of violence), and **Non-Violence** (peaceful content). The dataset is split into 2,700 train, 1,330 validation, and 2,016 test samples. All experiments are evaluated on the validation set for direct comparison with published baselines. To ensure consistency, we reran the saved BanglaBERT model checkpoint from the original work and obtained validation results closely matching those reported in the paper [7] (i.e. I already had the saved model).

4.2 Models and Prompting

We evaluate **GPT-4o** [5] under two conditions. For **zero-shot**, we provide a system prompt encoding the paper's adaptive decision framework [7] with precise category definitions, instructing the model to output a single digit (0, 1, or 2). For **few-shot**, we augment the prompt with 3 labeled examples per class (9 total) sampled from the training set, presented as alternating user/assistant turns. All experiments use temperature=0 for reproducibility. (Note, I wanted to add the prompts, but the 2 page limit, forced me to ommit the prompts picture from this report. I will include that in the final report.)

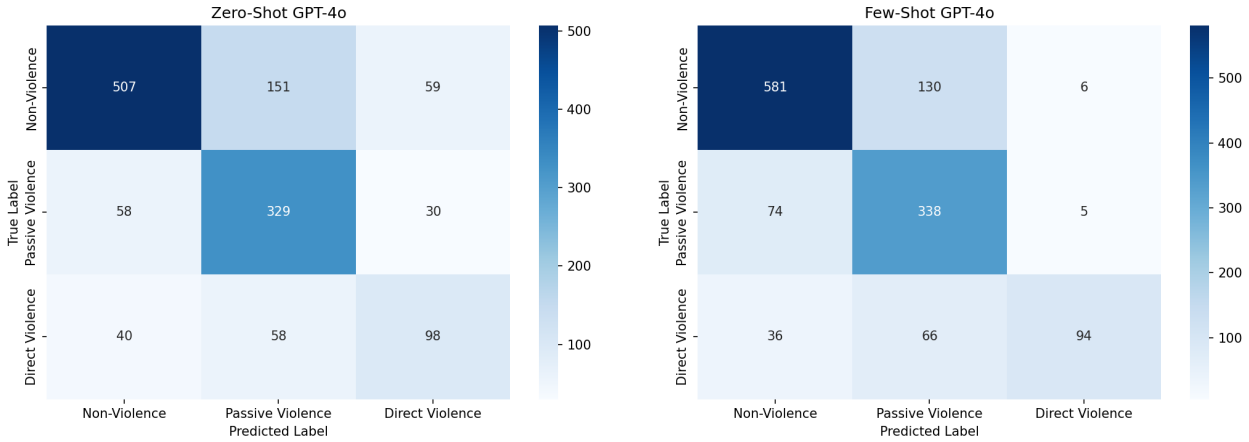

Figure 1: Confusion matrices for Zero-Shot (left) and Few-Shot (right) GPT-4o. Few-shot notably improves the overall result. Notably, providing in-context examples reduces extreme misclassifications, such as Direct Violence being predicted as Non-Violence and vice versa.

4.3 Baselines

We directly adopt published validation results from Saha et al. [7] for n-gram SVM baselines (Unigram, Char-2-gram, Char-3-gram) and fine-tuned transformer models (MBERT, XLM-RoBERTa, DistilBERT, BanglaBERT). Since we had access to the original saved model checkpoints from prior work, we verified these results by rerunning inference on the validation set, confirming consistency with the reported numbers.

4.4 Evaluation Metrics

We report per-class precision, recall, and F1-score, with macro-averaged F1 as the primary metric, consistent with prior work [7]. Macro F1 is appropriate given class imbalance: Direct Violence (14.4%), Passive Violence (34.1%), and Non-Violence (51.4%).

5 Preliminary Results

Table 1: Per-class and macro F1 on Vio-Lens validation set. Baseline results from [7].

Model	Direct	Passive	Non-V	Macro F1
N-gram Baselines				
Unigram SVM	0.444	0.629	0.776	0.616
Char-2-gram SVM	0.490	0.630	0.788	0.636
Char-3-gram SVM	0.505	0.662	0.803	0.657
Fine-tuned Transformers				
DistilBERT	0.548	0.637	0.771	0.652
MBERT	0.601	0.643	0.802	0.682
XLM-RoBERTa	0.670	0.704	0.813	0.729
BanglaBERT	0.754	0.764	0.846	**0.788**
LLM Prompting (This Work)				
Zero-Shot GPT-4o	0.510	0.690	0.770	0.656
Few-Shot GPT-4o	0.620	0.710	0.830	0.720

Table 1 presents our results. Zero-shot GPT-4o achieves a macro F1 of 0.6559, surpassing all n-gram SVM baselines without any training data, falling only 0.026 points below fine-tuned MBERT. Few-shot prompting yields a substantial improvement to macro F1 of 0.7202, exceeding MBERT (0.682) and approaching XLM-RoBERTa (0.729), both of which were fully trained on 2,700 labeled examples.

Per-class analysis reveals important patterns. The most notable gain from few-shot is on Direct Violence (+0.11 F1), the hardest and most imbalanced class. Few-shot Direct Violence precision reaches 0.90, indicating that when the model predicts this class it is highly accurate. However, recall remains low at 0.48, suggesting the model is still conservative in flagging direct violence. This mirrors the pattern observed across all baselines in prior work [7]. Non-Violence and Passive Violence both improve consistently under few-shot, with Non-Violence reaching F1=0.83, on par with fine-tuned transformer models.

These results confirm that LLMs are strong competitors for low-resource violence detection, with few-shot GPT-4o achieving near fine-tuned transformer performance using only 9 in-context examples.

6 Future Work

We plan to add **LLaMA 3 8B** [9] as a second LLM baseline, running locally on GPU infrastructure to enable cost-free evaluation and broader model comparison. We will also investigate robustness through three strategies: (1) **paraphrasing** violence-inciting texts to test classification stability under semantic-preserving transformations; (2) **word masking** of key violent terms to measure model sensitivity to specific lexical cues; and (3) **noise injection** via character substitutions and typos to simulate real-world social media noise.

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